

Chapter 12- Active Directory, OU, Group Policy & Server Practical

1. Active Directory (AD) – Complete Concept

◆ What is Active Directory?

Active Directory (AD) is a **directory service** developed by **Microsoft** used to manage:

- Users 
- Computers 
- Resources (printers, files, etc.) 

 It stores **complete employee data in one place**

What data is stored?

- Employee Name
 - Email ID 
 - Designation
 - Mobile Number 
 - Job Location 
 - Login Credentials 
-

Simple Meaning:

 Like a **company database + security system combined**

Key Features

1. Centralized Management

- Admin controls everything from **one place**
 - No need to go to each computer
-

2. Authentication

- Verifies:
 - Username
 - Password

 “Are you valid user?” 

3. Authorization

- Controls:
 - What user can access
 - What user cannot access
-

4. Group Policy

- Apply rules to multiple users/systems
-

5. Domain Management

- Organizes:
 - Users
 - Computers
-

Where used?

- Offices
 - Companies
 - Enterprise networks
-

Daily Life Example:

Like a **school system** 

- Principal = Admin
 - Students = Users
 - Classes = OU
 - Rules = Group Policy
-

2. Organizational Unit (OU)

◆ What is OU?

OU is a **container** inside Active Directory used to organize:

- Users
 - Groups
 - Computers
-

Why use OU?

- Easy management
 - Apply policies easily
-

Real Scenario (IMPORTANT)

Company structure:

- Departments:
 - IT
 - HR
 - Finance
 - Marketing
- Locations:
 - Mumbai
 - Pune
 - Nashik

Security Scenario:

 Suppose:

- Mumbai location is under attack 

Admin action:

- Convert users → **Standard User (not admin)**

Why?

- Admin → Full control (danger)
- Standard → Limited access (safe)

Smart OU Design:

- Create OU:
 - Mumbai
 - Inside Mumbai → Non-Tech OU

👉 Add:

- HR
- Finance
- Marketing

❌ Don't add IT users

🧪 Example:

Like **grouping students by class + subject** 🎓

⚙️ 3. Group Policy (GPO)

◆ What is GPO?

Group Policy is used to:

👉 Control user & system settings centrally

🔍 Features

● Centralized Control

- Apply settings to many systems at once
-

🔒 Security Control

- Password rules
 - Lock account
-

👤 User Configuration

- Desktop settings
 - Access control
-

Computer Configuration

- System updates
 - Security settings
-

Automation

- Runs automatically on:
 - Login
 - Startup
-

Real Example:

- Company wallpaper fixed 
- Cannot open Control Panel 

 Done using **GPO**

Technical Insight:

 GPO works using:

- Registry changes
 - Applied on OU
-

Example:

Like school rule:

 “Uniform is compulsory” 

4. PRACTICAL TASK – Server Setup + AD + GPO (STEP-BY-STEP)

STEP 1: Download Server OS ISO

1. Go to browser 
2. Search:

Windows Server ISO download

3. Download latest version
-

STEP 2: Install in VMware / VirtualBox

1. Open VMware
 2. Click → Create New VM
 3. Select ISO
 4. Choose:
 - OS: Windows Server
 5. Allocate:
 - RAM: 4GB
 - CPU: 2–4 cores
 6. Finish → Start VM
-

STEP 3: Install Windows Server

- Same as Windows installation
 - Set password for Administrator 
-

STEP 4: Install Active Directory

1. Open **Server Manager**
 2. Click:
 Add Roles and Features
 3. Select:
 Active Directory Domain Services
 4. Install
-

STEP 5: Promote to Domain Controller

1. Click:
 Promote this server
 2. Choose:
 Add new forest
 3. Enter domain:
company.local
 4. Set password
 5. Restart system
-

STEP 6: Create OU

1. Open:
 Active Directory Users and Computers
 2. Right-click domain
 3. Click → New → Organizational Unit
 4. Create:
 - Mumbai
 - Non-Tech
-

STEP 7: Create Users

1. Right-click OU
2. Click → New → User
3. Enter:
 - Name
 - Username
 - Email
 - Password

Enable:

✓ User must change password at next login

👉 This is called **Onboarding**

Onboarding Process

- Admin creates account
- Manager shares credentials
- User logs in → changes password

Offboarding Process

👉 When employee leaves:

1. Disable account (90 days)
 2. Then delete permanently
-

STEP 8: Create Group Policy

1. Open:
 - 👉 Group Policy Management
 2. Right-click OU
 3. Create GPO
-

Task 1: Disable Control Panel

Go to:

User Configuration → Administrative Templates → Control Panel

Enable:

- 👉 Prohibit access to Control Panel
-

Task 2: Set Wallpaper

Go to:

User Configuration → Administrative Templates → Desktop

Set:

- Wallpaper path
 - Enable policy
-

STEP 9: Link GPO to OU

- Right-click OU
 - Link existing GPO
-

STEP 10: Apply Policy

Run:

```
gpupdate /force
```

Final Result

- ✓ Control Panel disabled 
 - ✓ Wallpaper fixed 
 - ✓ Policy applied to selected users
-

IT SUPPORT + CYBER SECURITY INSIGHTS

Why AD is important?

- Central control
 - Security enforcement
 - Easy management
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Attack Prevention:

- Limit admin rights
 - Use OU-based control
 - Apply strict GPO
-

Troubleshooting Tips:

User cannot login:

- Check AD account
 - Reset password
-

✗ GPO not working:

- Run:

gpresult /r

✗ Policy not applied:

- Check OU link
 - Run gpupdate
-

Real-Life Summary

Concept	Example
Active Directory	Company database 
OU	Department group 
GPO	Company rules 
Onboarding	Joining process 
Offboarding	Exit process 

Q & A

? Q1: What is Active Directory?

✓ Answer:

Active Directory (AD) is a directory service developed by Microsoft that is used to centrally manage users, computers, and resources in a network environment.

👉 It stores:

- User accounts 
- Computer accounts 
- Passwords 
- Permissions & policies

💡 Simple Explanation:

It acts like a **central database + security system** of a company.

🎯 Interview Tip:

“Active Directory helps IT admins manage all users and systems from a single location instead of configuring each system manually.”

? Q2: What are the main features of Active Directory?

✓ Answer:

◆ 1. Centralized Management

Admin controls all users & systems from one place.

◆ 2. Authentication

Verifies username & password

👉 “Are you a valid user?”

◆ 3. Authorization

Controls what user can access

◆ 4. Group Policy

Apply rules to multiple users

◆ 5. Domain Management

Organizes entire network

? Q3: What is a Domain?

✓ Answer:

A domain is a logical group of computers, users, and resources managed under Active Directory.

💡 Example:
company.local

👉 All users login using:
username@company.local

? Q4: What is a Domain Controller?

✓ Answer:

A Domain Controller (DC) is a server that runs Active Directory and handles authentication & authorization.

👉 It verifies:

- Username
- Password

🎯 Without DC → No login in domain environment

ORGANIZATIONAL UNIT (OU) – INTERVIEW QUESTIONS

 **Q5: What is OU in Active Directory?**

 **Answer:**

OU (Organizational Unit) is a container used to organize users, groups, and computers.

 **Example:**

- IT Department
 - HR Department
 - Finance
-

 **Q6: Why do we use OU?**

 **Answer:**

- Easy management 
 - Apply policies easily 
 - Logical grouping 
-

 **Interview Line:**

“OU helps in applying policies department-wise instead of applying to entire organization.”

 **Q7: What is the best practice for OU design?**

 **Answer:**

 **Based on:**

- Department (HR, IT, Finance)
- Location (Mumbai, Pune)

Important:

- Separate IT users from normal users
 - Apply strict policies to non-tech users
-

Q8: What happens if OU is not used?

Answer:

- Difficult to manage users
 - Cannot apply targeted policies
 - Poor security control
-

GROUP POLICY (GPO) – INTERVIEW QUESTIONS

Q9: What is Group Policy?

Answer:

Group Policy is used to control user and computer settings centrally in Active Directory.

 It applies rules automatically.

Q10: What are types of Group Policy?

Answer:

User Configuration

- Desktop settings
- Restrictions

Computer Configuration

- Security settings
- Updates

? Q11: What is GPO used for in real company?

✓ Answer:

- Disable Control Panel 🚫
 - Set company wallpaper 🖼️
 - Restrict USB 🔌
 - Password policy 🔒
-

? Q12: How does GPO work internally?

✓ Answer:

GPO works by modifying Windows Registry settings automatically.

👉 Applied during:

- Login
 - Startup
-

? Q13: What is gpupdate /force?

✓ Answer:

Command used to force update Group Policy immediately.

👉 Without restart

? Q14: How to check applied policies?

✓ Answer:

Use command:

gpresult /r

👉 Shows applied GPOs



PRACTICAL INTERVIEW QUESTIONS (VERY IMPORTANT)

? Q15: How do you install Active Directory?



Answer (Step-by-Step):

1. Open Server Manager
 2. Click → Add Roles and Features
 3. Select:
 - ✓ Active Directory Domain Services
 4. Install
 5. Promote to Domain Controller
 6. Create new forest (example: company.local)
-

? Q16: How do you create a user in AD?



Answer:

1. Open AD Users & Computers
2. Right-click OU
3. New → User
4. Enter details
5. Set password

✓ Enable:

“User must change password at next login”

? Q17: What is Onboarding?

✓ Answer:

Process of creating user account and giving access.

👉 Steps:

- Create account
 - Assign permissions
 - User changes password
-

? Q18: What is Offboarding?

✓ Answer:

Process when employee leaves company.

👉 Steps:

1. Disable account (90 days)
2. Then delete

🎯 Prevents misuse

? Q19: How to apply GPO?

✓ Answer:

1. Open Group Policy Management
2. Create GPO
3. Link to OU
4. Run:

gpupdate /force

REAL SCENARIO QUESTIONS (MOST IMPORTANT)

 **Q20: User cannot login to domain. What will you do?**

 **Answer:**

Step-by-step approach:

1. Check username/password 
2. Check AD account status
3. Reset password
4. Check network connectivity 
5. Check Domain Controller

 “Always follow structured troubleshooting”

 **Q21: GPO not applying. What will you check?**

 **Answer:**

1. Check OU link
 2. Run:
`gpupdate /force`
 3. Run:
`gpresult /r`
 4. Check security filtering
 5. Restart system
-

? Q22: Why should we remove admin rights from users?

✓ Answer:

👉 Because admin has full control:

- Install software
- Disable security
- Run malware

🎯 Standard user = safer environment

? Q23: How to secure company using AD?

✓ Answer:

- Use OU structure 📁
 - Apply strict GPO 🔒
 - Disable admin rights
 - Use password policies
 - Monitor login logs
-

? Q24: What happens if Domain Controller fails?

✓ Answer:

- Users cannot login
- Authentication fails
- Network stops

👉 Solution:

- Use Backup Domain Controller
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ADVANCED INTERVIEW QUESTIONS

 **Q25: What is LDAP?**

 **Answer:**

LDAP is protocol used by AD to communicate with clients.

 **Q26: What is Kerberos?**

 **Answer:**

Authentication protocol used in AD.

 Secure login system 

 **Q27: What is SYSVOL folder?**

 **Answer:**

Stores Group Policy files.

 Shared across domain controllers

 **Q28: Difference between Authentication & Authorization?**

 **Answer:**

Type

Meaning

Authentication Who are you?

Authorization What can you access?



TROUBLESHOOTING (JOB-ORIENTED QUESTIONS)



Q29: User account locked. What will you do?



Answer:

- Unlock account in AD
 - Check failed login attempts
 - Reset password
-



Q30: New user not getting GPO?



Answer:

- Check OU placement
 - Run gpupdate
 - Restart system
-



Q31: System not joining domain?



Answer:

- Check network
 - Check DNS
 - Check domain name
 - Check credentials
-